

Written Exam — Functional Ecology in R

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Duration	Format	Resources	Total
60 min	Individual · in-class	Notes and exercises allowed	40 points → /20

Answer in English or French. No R code required. Write in the spaces provided. Be concise — 3 to 5 sentences per question is enough.

Question 1 (10 pts)

You computed FRic for 30 river basins and found a strong positive correlation with species richness ($r = 0.82$, $p < 0.001$). Your colleague concludes: *'Species-rich rivers are also functionally diverse.'*

a) Is this conclusion valid without further analysis? Explain why in 2–3 sentences. **(4 pts)**

b) What analysis should be done to test whether assemblages are *truly* more functionally diverse than expected? Name the metric and describe its logic in 2–3 sentences. **(6 pts)**

Question 2 (10 pts)

Two freshwater fish communities both have $S = 12$ species and CWM body length = 8 cm. Community 1 has FRic = 0.70. Community 2 has FRic = 0.15.

a) Explain in your own words what these FRic values tell us about the two communities. A small sketch of the trait space is welcome but not required. **(5 pts)**

b) Propose one ecological mechanism that could explain the low FRic of Community 2. **(5 pts)**

Question 3 (10 pts)

The figure below (hypothetical) shows SES-FR_{ic} values for fish assemblages across three biogeographic realms. The dashed lines mark ± 1.96 .

Realm	Median SES-FR _{ic}	Interpretation
Neotropical	+2.6	Most values above +1.96
Palearctic	-1.8	Several values below -1.96
Nearctic	0.1	Values scattered around 0

a) What assembly process does the Neotropical pattern (SES $\approx$ +2.6) suggest? Name one ecological mechanism that could drive it. **(5 pts)**

b) What assembly process does the Palearctic pattern (SES $\approx$ -1.8) suggest? Name one ecological mechanism. **(5 pts)**

Question 4 (10 pts)

In 4–6 sentences, explain why measuring **functional diversity** — rather than species richness alone — is important for biodiversity conservation. Use at least one example from freshwater fishes.

End of exam. Total: 40 pts → grade /20.